SANT'ANNA INSTITUTE SORRENTO

COURSE NUMBER AND TITLE: MATH 331 - Differential Equations (4 credits)
DOCUMENT: Lecture Plan, Personal Projects, and Open Textbook
Prerequisites: Calculus II

PURPOSE OF THIS DOCUMENT

This document turns the MATH 331 syllabus into an operational lecture plan. It covers (1) the prerequisite sessions that bring every student to the same starting point, (2) a session-by-session schedule that addresses all fifteen units, the seven learning outcomes, and the assessment scheme, with theoretical and practical exercises for each session, (3) five personal projects, and (4) an open textbook with additional readings that are freely and legally accessible online.

PLANNING ASSUMPTIONS

- The class meets twice a week for 2 hours over -- weeks, giving -- **sessions**. The comprehensive final exam takes place in the examination week.
- **Numerical thread.** Unit 14 is formally treated at the end of the course, but Euler's method is previewed in Week 4, and computational questions appear in every problem set, so that students can support their projects and Learning Outcome 6 from the start.
- **Applications thread.** Unit 15 is distributed throughout the course: circuits (Weeks 4 and 8), mechanical vibrations (Weeks 7 and 8), and population models (Weeks 4 and 10). Session 30 provides the synthesis.
- **Computational environment.** Python with Jupyter, numpy, scipy, and matplotlib, all free and open source. MATLAB or GNU Octave are acceptable alternatives.

ASSESSMENT ALIGNMENT

Component	Weight	Implementation in this plan
Homework Assignments	20%	12 weekly problem sets, each with analytical problems and one computational question.
Quizzes	10%	6 short quizzes (Sessions 5, 8, 12, 16, 20, 26), about 15 minutes each, on the most recent topics.
2 Mid Term Exams	30%	Midterm 1 (Session 10): Units 1-4. Midterm 2 (Session 22): Units 5-9. Each preceded by a review workshop.
Final Exam	25%	Comprehensive exam on Units 1-15 in the examination week.
Attendance and Participation	15%	Attendance, in-class problem solving, and the project presentation in Session 30.

PART 1 - PREREQUISITES: A COMMON STARTING POINT

The first week establishes the mathematical tools on which the course relies. An ungraded diagnostic quiz in Session 1 identifies gaps early; students who need further support receive targeted review exercises.

Session	Title	Content	Why it matters
1	Toolkit I: Calculus	Differentiation rules, implicit differentiation, partial derivatives; integration by substitution and by parts; partial fractions; improper integrals.	Separable and exact equations (Units 2-3), integrating factors, Laplace transforms (Units 10-11).
2	Toolkit II: Algebra and Computing	Complex numbers and Euler's formula; roots of polynomials; matrices, determinants, linear systems; Taylor series and radius of convergence; setup of Python and Jupyter.	Complex roots (Unit 5), systems (Units 8-9), series solutions (Units 12-13), numerical work (Unit 14).

Two short refreshers (about 20 minutes each) are placed just in time: eigenvalues and eigenvectors at the start of Session 17, and power series at the start of Session 27.

PART 2 - LECTURE SCHEDULE

Each session lists the syllabus unit, the topics, and one theoretical and one practical exercise. Theoretical exercises develop proofs, derivations, and conceptual understanding; practical exercises develop problem solving, modeling, and computation.

Wk	#	Unit	Topics	Theoretical exercise	Practical exercise
1	1	Prereq.	Toolkit I: calculus (see Part 1). Diagnostic quiz.	Derive the partial fraction decomposition of a rational function.	Integration drill: substitution, parts, partial fractions.
1	2	Prereq.	Toolkit II: complex numbers, linear algebra, series, Python setup.	Prove Euler's formula from Taylor series.	First Jupyter notebook: plot functions and solve a linear system.
2	3	U1	Definitions and terminology; order and degree; linear vs. nonlinear; verifying solutions; IVPs and BVPs; direction fields.	Classify ODEs by order, degree, and linearity; verify candidate solutions.	Plot slope fields in Python and sketch solution curves.
2	4	U2	Separable equations; implicit solutions.	Determine the domain of validity of a solution.	Torricelli's law: time to drain a tank.
3	5	U2	Linear first-order equations; integrating factor; existence and uniqueness theorems.	State the hypotheses of the theorem and find the interval of existence.	Analyze the non-unique case y' = y^(1/3). **Quiz 1.**
3	6	U3	Exact equations; integrating factors; Bernoulli and homogeneous substitutions.	Prove the test for exactness.	Find integrating factors of the form mu(x) or mu(y).

Wk	#	Unit	Topics	Theoretical exercise	Practical exercise
4	7	U4	Growth and decay; Newton's law of cooling; mixing tanks; logistic equation; autonomous equations and the phase line.	Classify the stability of equilibria.	Build a mixing-tank model from a verbal description.
4	8	U4/U15	Falling bodies with drag; RC and RL circuits; preview of Euler's method.	Derive the terminal velocity.	Compare Euler's method with the exact solution. **Quiz 2.**
Week 5 - Session 9: Review workshop on Units 1-4. Session 10: MIDTERM EXAM 1 (Units 1-4).					
6	11	U5	Linear independence; Wronskian; superposition; characteristic equation with distinct real roots.	Apply Abel's theorem.	Solve initial-value problems.
6	12	U5	Complex and repeated roots; general solutions.	Derive the e^(at)cos(bt) solution form from Euler's formula.	Solve IVPs and sketch the solutions. **Quiz 3.**
7	13	U6	Method of undetermined coefficients.	Explain when the trial solution must be modified.	Solve equations with a variety of forcing terms.
7	14	U6/U15	Variation of parameters; free and damped vibrations.	Derive the variation-of-parameters formula.	Classify a spring-mass system by its damping.
8	15	U6/U15	Forced vibrations; beats; resonance; RLC circuits.	Establish the mechanical-electrical analogy.	Plot an amplitude-response curve.
8	16	U7	Theory of nth-order equations; reduction of order; higher-order constant-coefficient equations; two-point BVPs (beam deflection).	Show that a BVP can have no solution or infinitely many solutions.	Compute the deflection of a beam. **Quiz 4.**
9	17	U8	Eigenvalue refresher; rewriting an nth-order ODE as a system; matrix form.	Prove the equivalence of the conversion to a first-order system.	Model a pair of coupled tanks.
9	18	U8	Eigenvalue method for real, complex, and repeated eigenvalues; fundamental matrix.	Find generalized eigenvectors.	Solve a two-mass oscillator.
10	19	U9	Phase plane; equilibria; nodes, saddles, spirals, and centers; trace-determinant plane.	Classify linear systems from their trace and determinant.	Draw phase portraits in Python.
10	20	U9/U15	Nonlinear systems; Jacobian linearization; stability; predator-prey, competing species, pendulum.	Explain when linearization predicts stability correctly.	Analyze the Lotka-Volterra model. **Quiz 5.**
Week 11 - Session 21: Review workshop on Units 5-9. Session 22: MIDTERM EXAM 2 (Units 5-9).					
12	23	U10	Laplace transform: definition, existence, properties, table; inverse transforms by partial fractions.	Prove the derivative rule.	Compute transforms and inverse transforms.

Wk	#	Unit	Topics	Theoretical exercise	Practical exercise
12	24	U10	Solving initial-value problems with the Laplace transform.	Compare the method with undetermined coefficients.	Solve an RLC circuit IVP.
13	25	U11	Heaviside step functions; shift theorems; discontinuous forcing.	Prove the second shift theorem.	Model a switched circuit.
13	26	U11	Dirac delta; impulse response; convolution; transfer functions.	Express the solution through the convolution theorem.	Model a hammer blow on a spring-mass system. **Quiz 6.**
14	27	U12	Power series refresher; series solutions at ordinary points; recurrence relations (Airy, Hermite).	Bound the radius of convergence from the singularities.	Compute the first coefficients numerically.
14	28	U13	Regular singular points; Euler equations; Frobenius method; Bessel functions.	Discuss the indicial equation and its three root cases.	Compare a series solution with the scipy Bessel functions.
15	29	U14	Euler, improved Euler (Heun), and fourth-order Runge-Kutta methods; local and global error; order of accuracy; stiffness; scipy solve_ivp.	Derive the order of accuracy of Euler's method.	Convergence study on a log-log plot.
15	30	U15	Course synthesis; project presentations; final review.	Connect each application to the method that solves it.	Ten-minute project presentations.
Examination week: FINAL EXAM (comprehensive, Units 1-15).					

Coverage of the Learning Outcomes

Learning outcome (syllabus)	Sessions and activities
1. Solve first- and higher-order ODEs using appropriate analytical methods	Sessions 4-6, 11-16
2. Analyze IVPs and BVPs; existence, uniqueness, and behavior of solutions	Sessions 3, 5, 11, 16
3. Apply systems of ODEs and linear algebra to dynamic systems	Sessions 17-20
4. Analyze ODEs with Laplace transforms, power series, and other methods	Sessions 23-28
5. Model and solve real-world problems and interpret the solutions	Sessions 7-8, 14-15, 20, 24-26; personal projects
6. Apply computational tools and numerical methods	Sessions 2, 3, 8, 19, 29; every problem set; personal projects
7. Communicate mathematical reasoning with notation, graphs, and terminology	Written proofs in homework and exams; project report and presentation

PART 3 - PERSONAL PROJECTS

Each student completes one individual project during the course. Every project combines an analytical part, a numerical part, and a physical or biological interpretation of the results.

Timeline and Deliverables

Week	Milestone
5	Projects presented and assigned.
7	One-page proposal: model, questions to answer, planned methods.
11	Checkpoint: analytical results and a working numerical code.
15	Final report (6-10 pages) with commented code, and a ten-minute presentation in Session 30.

Suggested Evaluation Rubric

Criterion	Weight	Description
Analytical work	30%	Correct derivation of the model and of the closed-form or qualitative results.
Numerical work	30%	Correct, verified, and reproducible code; sound choice of methods and step sizes.
Interpretation	20%	Physical or biological meaning of the results; limits of the model.
Communication	20%	Clarity of the report, figures, notation, and presentation.

Project 1 - Epidemics with the SIR Model *(Units 4, 9, 14)*

- Formulate the SIR model and find its equilibria and the threshold quantity R0.
- Explain analytically why the epidemic peaks and when it dies out.
- Integrate the model numerically and study how vaccination or a reduced contact rate changes the peak.
- Extension: add births and deaths (vital dynamics) and discuss the new equilibrium.

Project 2 - Coupled Oscillators and a Tuned Mass Damper *(Units 6, 8, 15)*

- Model a two- or three-story building as a chain of masses and springs.
- Compute the normal modes and natural frequencies from the eigenvalue problem.
- Compute the forced response near resonance under harmonic ground motion.
- Design a tuned mass damper that reduces the resonant peak and quantify its effect.

Project 3 - The Nonlinear Pendulum *(Units 5, 9, 14)*

- Compare the small-angle linear model with the full nonlinear model.
- Draw the phase portrait, identify the separatrix, and examine conservation of energy.
- Compute how the period depends on the amplitude.

- Add damping and periodic forcing and describe qualitatively the irregular motion that appears.

Project 4 - Switched RLC Circuit with Laplace Transforms *(Units 10, 11)*

- Model an RLC circuit driven by a step, a square wave, and an impulse.
- Solve analytically with transfer functions and convolution.
- Verify the analytical results against a numerical solution.
- Plot the frequency response and relate it to the damping ratio.

Project 5 - The Hanging Chain and Bessel Functions *(Units 12, 13)*

- Derive the equation for the small oscillations of a hanging chain.
- Reduce it to Bessel's equation and solve it with the Frobenius method.
- Compute the natural frequencies from the zeros of J0.
- Compare the results with a finite-difference numerical model.

OPEN TEXTBOOK AND ADDITIONAL READINGS

Main Resource

Jiri Lebl, Notes on Diffy Qs: Differential Equations for Engineers. Free under a Creative Commons license at https://libretexts.org/, with an additional LibreTexts edition. The book is adopted by many universities and was written to cover the same material as Boyce and DiPrima, the reference text named in the syllabus. Its chapters on first-order equations (including exact equations and numerical methods), higher-order linear equations, systems, Laplace transforms, power series and the Frobenius method, and nonlinear systems map closely onto Units 1-15.

Additional Readings

- **William F. Trench, Elementary Differential Equations with Boundary Value Problems.** Free from the Trinity University Digital Commons. A rich source of additional worked problems, especially for Units 3, 12, and 13.
- **MIT OpenCourseWare 18.03SC, Differential Equations** (A. Mattuck, H. Miller, and others). Lecture videos, notes, problem sets with solutions, and interactive applets: https://ocw.mit.edu/courses/18-03sc-differential-equations-fall-2011/
- **Learn Differential Equations: Up Close with Gilbert Strang and Cleve Moler** (MIT OpenCourseWare video series). Recommended for Units 8, 9, and 14.
- **For the prerequisite sessions:** OpenStax Calculus, Volume 2, for integration techniques and series (https://openstax.org/details/books/calculus-volume-2), and Stephen Boyd and Lieven Vandenberghe, Introduction to Applied Linear Algebra, for matrices and eigenvalues (https://web.stanford.edu/~boyd/vmls/).

INSTRUCTOR NOTES ON THE SYLLABUS

- **Project weight.** The personal projects are not part of the current assessment scheme. One option is to split Homework Assignments (20%) into Homework (10%) and Personal Project (10%).
- **Credits and contact hours.** The course is listed as 4 credits, whereas two 1.5-hour meetings per week correspond to 3 contact hours.
- **Inherited passages.** The personal technology section refers to literary discussions and paper topics, and the academic honesty section cites Jacksonville University rather than the Sant'Anna Institute.